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Portfolio:
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Education:

02/2025 – 06/2025
Bootcamp at Wbs Coding School
Full Stack Web & App Development
Focused on React, Node.js, JavaScript, and modern web development.

10/2009 – 02/2013
University of Technology in Kielce (Poland)
Bachelor of Science in Electrotechnics

Skills:

- MS Office, Git, Bitbucket, Jira
- JavaScript, Node.js, Express, React
- CSS, TailwindCSS, HTML5
- SQL, MongoDB, PostgreSQL, RestAPI, Postman
- Networking and general IT knowledge
- Python, Figma
- Familiar with Docker
- Tia Portal/TwinCat3 (PLC programming)
- WinCC/TwinCat HMI (Visualization)

Soft skills:

- **Problem-Solving**: Analytical mindset with a proven ability to troubleshoot complex automation and software issues under tight deadlines
- **Team Collaboration**: Experienced in working with cross-functional teams to align technical solutions with business objectives.
- **Adaptability**: Quickly learned and applied new technologies (e.g., React, Node.js, JS) during career transition to software development
- **Communication**: Effectively communicated technical concepts to operators, engineers, and stakeholders in multilingual environments
- **Time Management**: Successfully managed multiple projects and deadlines in fast-paced manufacturing and development settings

Languages:

English: C1
German: B2
Polish: Native

Adrian Bakalarz

Full Stack Developer

Controls Engineer transitioning to Full Stack/Software Developer with 6 years of experience in industrial automation, programming SCADA, HMI, and PLC systems, and 3 years leveraging Python and PostgreSQL at Tesla to optimize manufacturing processes. Analytical thinker and fast learner, adept at applying automation principles—where PLCs, motors, and sensors act as the backend and SCADA/HMI as the frontend/UI—to build scalable web applications. Building proficiency expertise in React and Node.js to deliver robust, user-focused software solutions.

Experience:

10/2021 – 06/2024 **Controls Engineer,** Tesla, Grunheide

- Updated and improved HMIs for complex machinery to be used by operators and maintenance.
- Implemented code modifications to PLCs to introduce new functionalities, adhering to Object Oriented Programming (OOP) principles. Tracked all changes efficiently using Git for version control.
- Supported commissioning in Berlin and Austin, ensuring seamless system integration.
- Collaborated with cross-functional teams to align control algorithms with production goals.
- Developed Python scripts to analyze sensor data and optimize control algorithms, reducing battery transport crane cycle time from 35 to 30 seconds (14.3% improvement).
- Utilized PostgreSQL to store and query historical process data, enabling rapid fault resolution.
- Designed PostgreSQL queries to monitor equipment performance, collaborating with teams to enhance predictive maintenance.
- Built Python tools for control logic simulation, interfacing with PostgreSQL to validate data integrity.

09/2018 – 08/2021 **PLC Programmer,** Simtech AP, Krakow (Poland)

- Programmed PLCs (e.g., Siemens Tia Portal) to control machinery, optimizing processes such as material handling and production line in automotive manufacturing, analogous to backend logic development in software applications.
- Developed intuitive SCADA and HMI interfaces for real-time process visualization and operator interaction, mirroring frontend/UI design principles in web development.
- Conducted system diagnostics and debugging of PLC code, resolving automation faults under tight deadlines, improving system uptime.
- Created detailed user guides and maintenance protocols for SCADA and HMI systems, enhancing operator training and system maintainability, comparable to writing technical specs for software projects.
- Documented code changes and system configurations, maintaining clear records for future upgrades, aligning with version control practices like Git in software development.